LearnLens AI × IBM Bob 2.0
Autonomous Multimodal Knowledge Ingestion & Intelligent Enterprise Learning Agent
Track: Enterprise SDLC & Autonomous Developer Systems | Status: LIVE IN PRODUCTION

CORE VALUE PROPOSITION

Show your AI what you are learning. LearnLens AI passively observes user-authorized Chrome tabs,
extracts genuine 16:9 lecture slides via perceptual difference hashing, transcribes speech in real
time, and auto-generates presentation decks and interactive quizzes.

OFFICIAL DEPLOYMENT LINKS

• Live Web App: https://transcript-agent-frontend.onrender.com/
• GitHub Repo: https://github.com/soham-092510/Transcript-Agent
• FastAPI Swagger Docs: https://transcript-agent-backend.onrender.com/docs
• System Health: https://transcript-agent-backend.onrender.com/api/system/status

TEAM MEMBERS & CREDENTIALS

Soham Haridas Gaikwad — Team Leader | AI, Developer & Innovation Sardar Patel Institute of Technology (SPIT)
GitHub: github.com/soham-092510 | LinkedIn: in/soham-gaikwad-0207b4380

Harshvardhan Sanjay Jangam — Systems & Backend Engineer
Shivajirao S. Jondhle College of Engineering & Technology (SSJCET)
GitHub: github.com/harshu0499 | LinkedIn: in/harshvardhan-jangam-a4681a2a5

Chinmay Tawade — UI & Full-Stack Developer | SPIT
Arnav Nandkishor Kinhekar — Data & Cloud Architect | DJ Sanghvi

LearnLens AI × IBM Bob 2.0 Hackathon Live: https://transcript-agent-frontend.onrender.com Slide 1 of 10

The Enterprise Knowledge & Training Bottleneck
Why Digital Upskilling and Technical Video Learning is Broken in Modern Organizations

1. The 100+ Hour Video Black Hole

Engineers spend 40–100 hours watching video lectures, workshops,
and architecture walkthroughs.

80% of critical knowledge (diagrams, terminal commands,
equations) is locked inside video pixels with zero searchability.

2. Cognitive Context Switching

Constantly pausing video to take manual notes and screenshot slides causes a 40% drop in concept retention.

Manual screenshots produce disorganized, blurry folders with no timestamp grounding or text search.

3. Mid-Lifecycle Knowledge Loss

Teams modernizing enterprise codebases lack verified study
materials or instant Q&A; tied to original lecture recordings.

Standard AI tools dump raw transcripts that completely ignore
visual slide context.

KEY IMPACT METRIC: 68% of technical learners report inability to quickly find visual diagrams shown in past video sessions. Video audio has zero text searchability without heavy cloud GPU indexing.

LearnLens AI × IBM Bob 2.0 Hackathon Live: https://transcript-agent-frontend.onrender.com Slide 2 of 10

LearnLens AI — The Autonomous Learning Companion
Passive Screen Observation Meets Active Multimodal Intelligence

1. Sub-50ms Chrome Tab Observation
Instantaneous screen capture picker without browser gesture timeouts. Seamlessly captures high-res
video and tab lecture audio directly from Coursera, YouTube, Udemy, and enterprise webinars.

2. Perceptual Keyframe Extraction
Client-side 16x9 difference hashing + OpenCV dHash Hamming distance. Captures only genuine visual slide changes with zero duplicates, eliminating 85% of redundant bandwidth.

3. Instant 16:9 Presentation Synthesis
One-click synthesis of clean, professional 16:9 widescreen PowerPoint (.pptx) decks matching exact
video dimensions, alongside publication-grade ReportLab PDF study packs.

4. Grounded AI Teacher & Practice Quizzes
Interactive doubt-solving teacher quoting exact timestamps and slide citations. Automatically generates
multiple-choice practice exams with instant grading and evidence links.

LearnLens AI × IBM Bob 2.0 Hackathon Live: https://transcript-agent-frontend.onrender.com Slide 3 of 10

Technical Architecture & Engineering Flow
Full-Duplex Multimodal Pipeline Orchestrated End-to-End with IBM Bob 2.0

FRONTEND CLIENT LAYER

• React 19 + TypeScript + Vite 8
• WebRTC getDisplayMedia capture
• Web Audio API 16kHz downsampler
• Web Speech API native STT
• In-memory 16x9 visual diff canvas

BACKEND STREAMING CORE

• FastAPI async server (Python 3.10)
• Full-duplex WebSocket bus
• Faster-Whisper (int8, VAD filter)
• OpenCV perceptual dHash engine
• Tesseract OCR with Semaphore(2)

SYNTHESIS & DATA PERSISTENCE

• SQLite 3 with WAL Journal Mode
• Compound B-tree indexed schema
• python-pptx 16:9 slide generator
• ReportLab vector PDF builder
• Ollama Qwen 2.5 grounded RAG

IBM BOB 2.0 ORCHESTRATION IN SDLC:

• Plan Mode: Architected the 3-tier streaming protocol, database migration schemas, and async queue boundaries.
• Ask Mode: Solved Chrome transient activation timing, Web Audio RMS gating (rms > 0.004), and CDN export URLs.
• Agent Mode: Automated backend routes, test suites (13 unit/integration tests with 100% pass), and Render YAML spec.

LearnLens AI × IBM Bob 2.0 Hackathon Live: https://transcript-agent-frontend.onrender.com Slide 4 of 10

Live Interface & Observation Studio (Production Showcase)
Enterprise-Grade Dark Mode Interface Engineered for Distraction-Free Learning

[PIN 1] Sub-50ms Tab Observation
Instantaneous WebRTC media stream with picture-in-picture instructor toggle. Zero lag or freezing.

[PIN 3] Real-Time Dual STT Ticker
Sub-second speech recognition with live word-by-word streaming ticker and speaker labeling.

[16:9 METALLIC LAPTOP INTERFACE CONTAINER]

LIVE WEB APPLICATION RUNNING AT: https://transcript-agent-frontend.onrender.com/

• Video Feed (Left): Real-time educational video observation with controls & speed adjust
• Keyframe Reel (Bottom): Chronological slide carousel with OCR annotations & pin toggles
• Live Transcript (Right): Live minute-bucketed transcripts with concept extraction
• Top Actions Bar: Instant 16:9 Slide PDF & PPTX Download, Teacher Chat, & Auto-Pilot

[PIN 2] Perceptual dHash Slides
Slide change carousel capturing verified keyframes with zero stationary duplicates.

[PIN 4] 1-Click Presentation Export
Download custom 16:9 PPTX decks and ReportLab study reports immediately with zero cloud delay.

LearnLens AI × IBM Bob 2.0 Hackathon Live: https://transcript-agent-frontend.onrender.com Slide 5 of 10

Autonomous HyperIngest Studio (65x Accelerated Mode)
Completing 11-Hour / 100-Video Master Courses Autonomously in 10-15 Minutes

AUTONOMOUS WORKFLOW
• Batch Video Directory Ingest: Drag and drop multi-part video playlists or complete courses.

• Non-Linear Keyframe Scanning: Multi-threaded frame evaluation scans 100+ lectures concurrently.

• Chrome Tab Auto-Pilot & Turbo Script: Integrated 1-click script runs videos at 2x–16x speed and
auto-clicks 'Next Lecture' on Coursera, Udemy, and YouTube.

• Instant Master Slide Deck: Compiles an entire course into a single unified 16:9 PowerPoint deck.

PRODUCTION BENCHMARKS
■ 65x REALTIME INGESTION

Ingests an 11-hour lecture series in under 12 minutes with full visual keyframe coverage. ■ 85% NETWORK REDUCTION

Client-side thumbnail diffing suppresses static frames, saving bandwidth on cloud instances. ■ 100% LOCAL PROCESSING

Zero raw video files transmitted to third-party APIs. Complete enterprise data privacy.

LearnLens AI × IBM Bob 2.0 Hackathon Live: https://transcript-agent-frontend.onrender.com Slide 6 of 10

Grounded AI Teacher & Practice Exam Engine
Eliminating Hallucinations Through Exact Slide and Timestamp Citations

ZERO-HALLUCINATION RAG

Every synthesized answer is bound directly to captured lecture
transcripts and OCR text.

The agent quotes the exact timestamp (e.g., [04:18]) and displays
the corresponding slide thumbnail as clickable evidence.

THREE PEDAGOGICAL MODES

• Simple Mode: Plain-English explanations with intuitive analogies.

• Deep Mode: Technical architectural breakdowns, formulas, and code mechanics.

• Exam Prep Mode: High-yield definitions, formula recap, and common exam traps.

DYNAMIC PRACTICE QUIZZES

Automatically generates multiple-choice practice exams tailored to
captured concepts.

Provides instant grading, comprehensive explanations, and lecture
timestamps for every question.

LearnLens AI × IBM Bob 2.0 Hackathon Live: https://transcript-agent-frontend.onrender.com Slide 7 of 10

Production Engineering, Feasibility & Cloud Scalability
High-Throughput System Optimization for Scalable Multi-User Deployments

1. Sub-Millisecond SQLite Queries
Enabled WAL (Write-Ahead Logging) and created compound B-tree indexes across all foreign keys
(session_id, timestamp_sec). Eliminates full-table scans across 500+ records, executing in <0.2ms.

2. Queue Purging & CPU Safeguards
Thread-capped Whisper inference (cpu_threads=1, num_workers=1) with automatic backlog queue purging. Guarantees zero audio queue lag and eliminates CPU lockup on cloud free tiers.

3. Client-Side RMS Silence Gating

Client Web Audio node discards audio chunks below RMS 0.004. Pauses redundant server audio
streaming whenever client Web Speech STT is active, saving 80% server compute.

4. Cross-Origin CDN Asset Resolution
Implemented dynamic getExportDownloadUrl helper with Content-Disposition headers and extension fallback matching, resolving 404 download errors on Render static site CDNs.

LearnLens AI × IBM Bob 2.0 Hackathon Live: https://transcript-agent-frontend.onrender.com Slide 8 of 10

Project Milestones & Development Timeline
The IBM Bob 2.0 Accelerated Delivery Roadmap (From Concept to Production in 4 Days)

DAY 1: PROTOTYPE & SPEC

• Architected multi-tier streaming protocol with
IBM Bob Plan Mode.
• Established WebRTC capture & database
schema.
• Verified base WebSocket connection.

DAY 2: VISION & STT ENGINE

• Implemented OpenCV dHash perceptual
difference algorithms.
• Configured Faster-Whisper audio workers.
• Integrated ReportLab & python-pptx generators.

DAY 3: HYPERINGEST & RAG

• Built autonomous batch video ingestion pipeline.
• Added 1-click Auto-Next console turbo script.
• Developed grounded RAG doubt-solving
teacher.

DAY 4: CLOUD & HARDENING

• Deployed full stack on Render via render.yaml.
• Resolved transient activation & CDN 404s.
• 100% test pass rate across 13 test suites.

LearnLens AI × IBM Bob 2.0 Hackathon Live: https://transcript-agent-frontend.onrender.com Slide 9 of 10

Transforming Digital Education at Enterprise Scale
The Future of Multimodal Knowledge Ingestion is Live and Open Source

CORE IMPACT & VALUE

• Converts passive video streaming into verified, structured presentations.

• Saves learners 60%+ of study time while maximizing retention.

• 100% open-source, local-first, privacy-preserving enterprise architecture.

• Future Roadmap: Chrome Web Store Extension, Canvas/Moodle LMS LTI integration, and multilingual translation.

OFFICIAL EVALUATION & DEMO LINKS

■ Live Deployed Application:
https://transcript-agent-frontend.onrender.com/

■ GitHub Repository:
https://github.com/soham-092510/Transcript-Agent

■■ FastAPI Interactive Docs:
https://transcript-agent-backend.onrender.com/docs

■ Health & Diagnostics:
https://transcript-agent-backend.onrender.com/api/system/status

Team: Soham Gaikwad (Leader) | Harshvardhan Jangam | Chinmay Tawade | Arnav Kinhekar

LearnLens AI × IBM Bob 2.0 Hackathon Live: https://transcript-agent-frontend.onrender.com Slide 10 of 10